

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Data Visualization with Power BI and Tableau

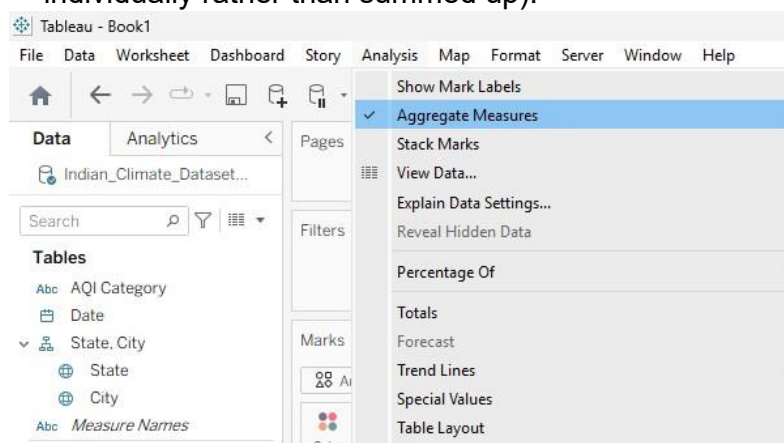
Practical - VIII

Roll No.: T031	Name: Ishita
Class: TYIT	Batch: 2
Date of Assignment: 27-8-26	Date/Time of Submission: 27-8-26

A: Add and Interpret Trend Lines (Linear, Exponential) :

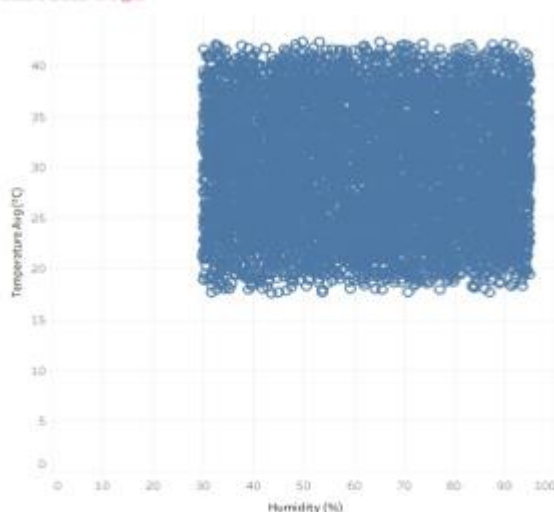
1. Connect Dataset & Build a Scatter Plot / Line Chart

1. Open Tableau Desktop and load Indian_Climate_Dataset_2024_2025.csv.
2. Open a new worksheet and rename it to Temperature vs Humidity Trends.
3. Drag Humidity (%) to the Columns shelf.
4. Drag Temperature_Avg (°C) to the Rows shelf.
5. In the top menu, go to Analysis and uncheck Aggregate Measures (so each data point is plotted individually rather than summed up).

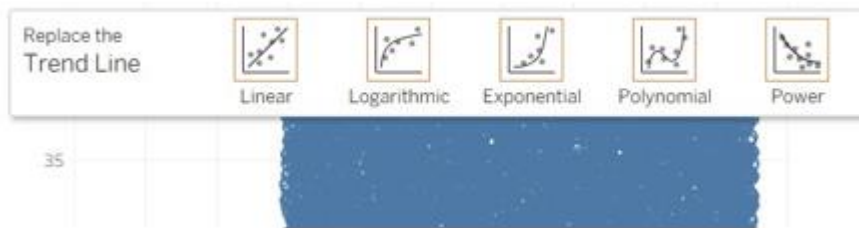


a Linear Trend Line

1. Switch to the Analytics pane on the left sidebar (tab next to Data).

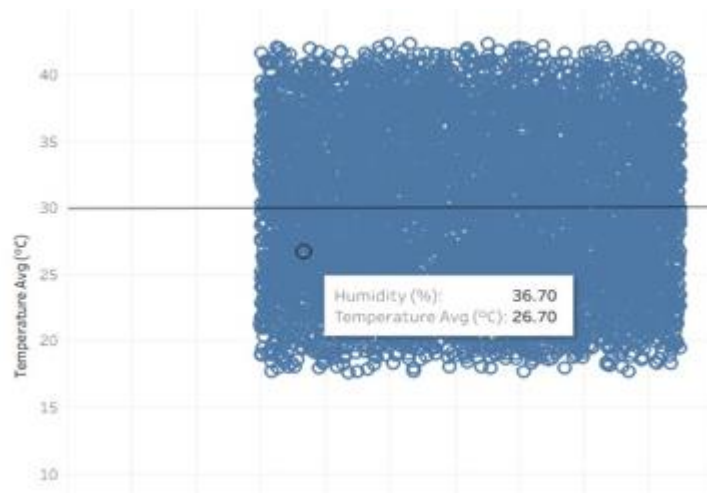


2. Under Model, drag Trend Line onto the canvas and drop it onto Linear.



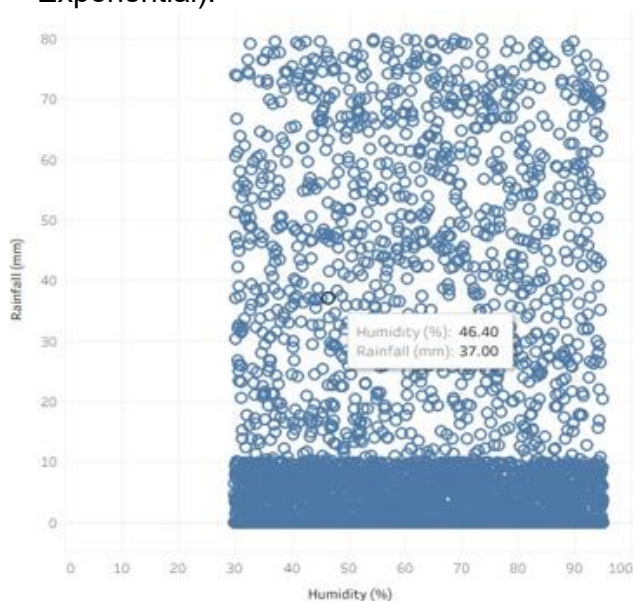
3. Hover over the trend line to observe the statistical popup:

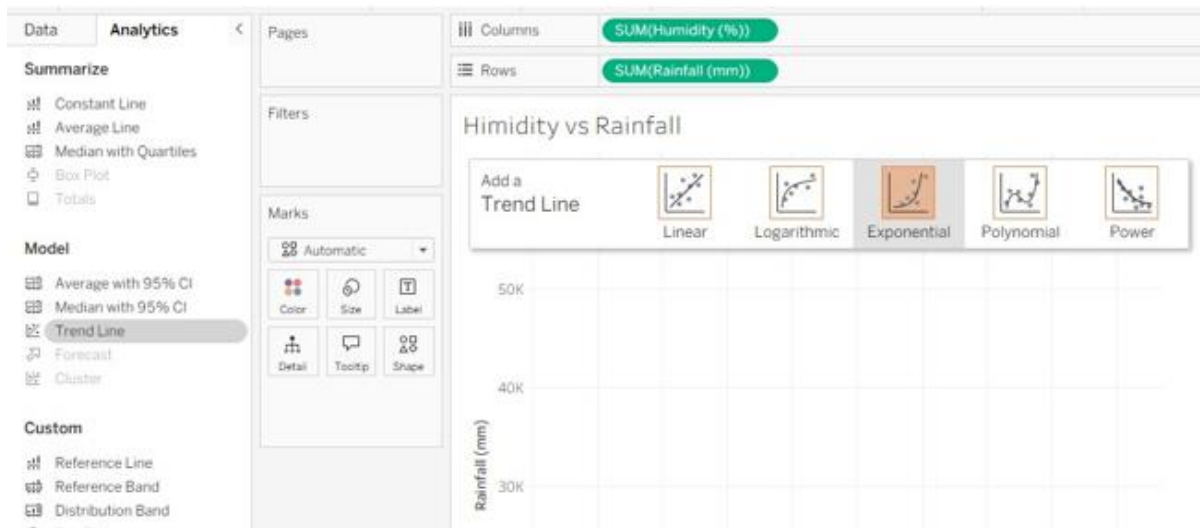
- It displays the linear equation: $\text{Temperature_Avg} = a \times \text{Humidity} + b$
- It displays R^2 (coefficient of determination) and P-value (statistical significance).



3. Add an Exponential Trend Line for Non-Linear Measures (Rainfall vs. Humidity)

- Open a new worksheet named Humidity vs Rainfall.
- Drag Humidity (%) to Columns and Rainfall (mm) to Rows.
- Uncheck Analysis ☐ Aggregate Measures.
- From the Analytics pane, drag Trend Line onto Exponential (or right-click the canvas ☐ Trend Lines ☐ Show Trend Lines ☐ Edit Trend Lines and change the model to Exponential).





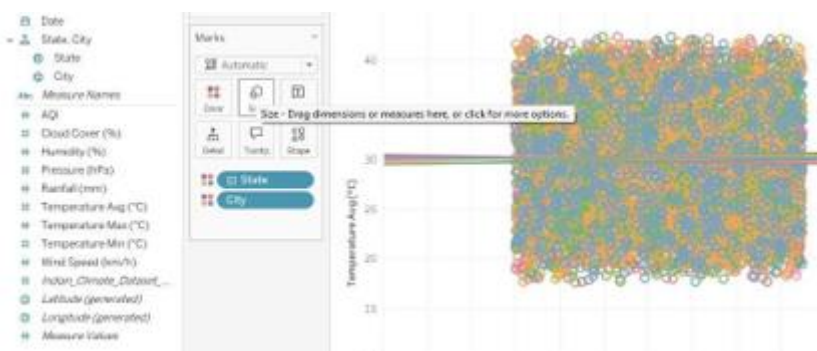
4. Interpret the Results

- Linear Model: Describes direct proportional/inverse relationships (e.g., checking if higher humidity steadily decreases or increases average temperature).
- Exponential Model: Captures accelerating relationships (e.g., how rainfall spikes exponentially once humidity crosses higher thresholds).

B: Customize and Analyze Trend Models :

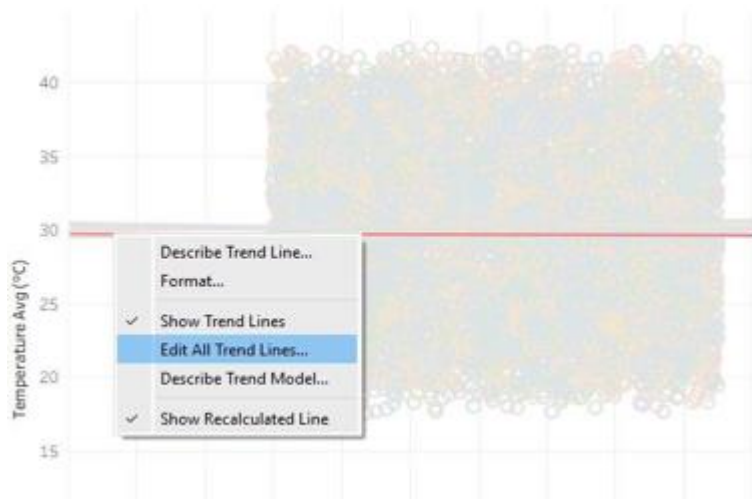
1. Disaggregate Trend Lines by City

1. Open your Temperature vs Humidity Trends sheet from Part A (or create a duplicate).
2. Drag City from the Data pane onto Color on the Marks card.
 - Notice what happens: Instead of one global trend line, Tableau automatically computes 10 distinct trend lines, one for each city!
3. Observe how coastal cities (like Mumbai or Chennai) show different slopes compared to inland cities (like Delhi or Jaipur).

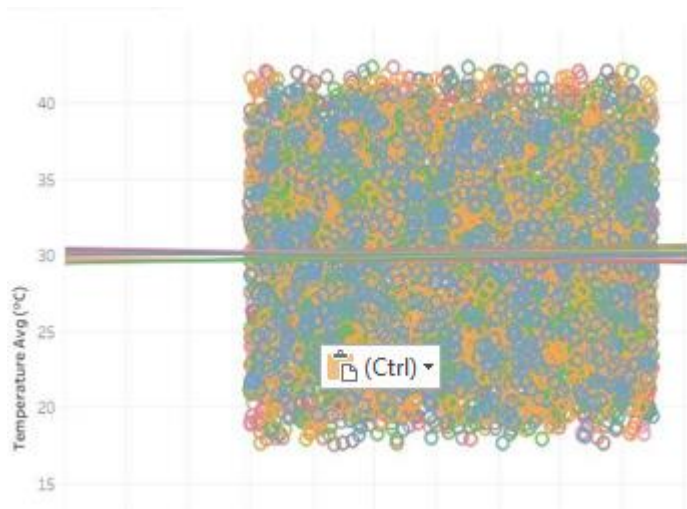


2. Compare Different Model Types (Polynomial & Logarithmic)

1. Right-click on any trend line in the view and select Edit Trend Lines...



2. In the options dialog, experiment with switching the Model Type:
 - o Linear: Assumes a constant rate of change.
 - o Logarithmic: Useful when values rise rapidly at first and then level off.
 - o Polynomial (Degree 2 or 3): Great for U-shaped or S-shaped seasonal curves (e.g., temperatures dipping mid-year and rising again).
3. Select Polynomial (Degree 2) and click OK to observe how well a curved line fits seasonal fluctuations.



X

Trend Lines Options

Model Type

☒ Linear
☐ Logarithmic
☐ Exponential
☐ Power
☐ Polynomial Degree: 3

Factors

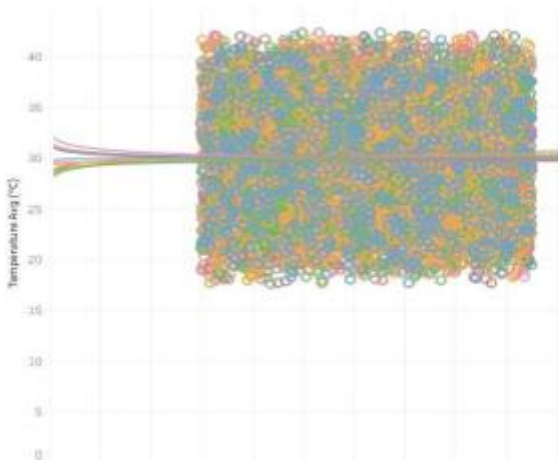
Build separate trend lines based on the following dimensions:

☒ City
☒ State

Options

☒ Show tooltips
☐ Show confidence bands
☒ Allow a trend line per color
☒ Show recalculated line for highlighted or selected data points
☐ Force y-intercept to zero

OK



Trend Lines Options

Model Type

☐ Linear

☒ Logarithmic

☐ Exponential

☐ Power

☐ Polynomial Degree: 3

Factors

Build separate trend lines based on the following dimensions:

☒ City

☒ State

Options

☒ Show tooltips

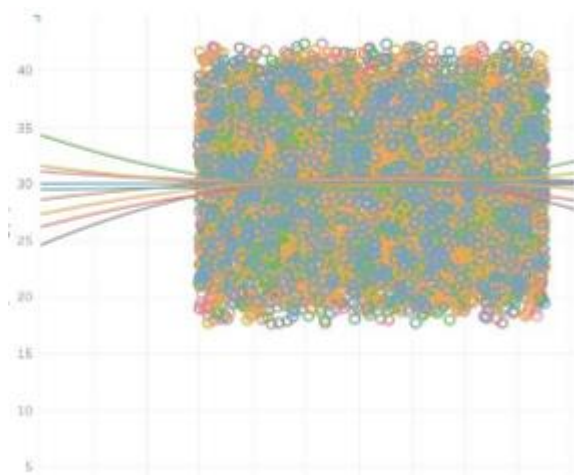
☐ Show confidence bands

☒ Allow a trend line per color

☒ Show recalculated line for highlighted or selected data points

☐ Force y-intercept to zero

OK



Trend Lines Options

Model Type

☐ Linear

☐ Logarithmic

☐ Exponential

☐ Power

☒ Polynomial Degree: 2

Factors

Build separate trend lines based on the following dimensions:

☒ City

☒ State

Options

☒ Show tooltips

☐ Show confidence bands

☒ Allow a trend line per color

☒ Show recalculated line for highlighted or selected data points

☐ Force y-intercept to zero

OK

3. Access Full Statistical Summaries (Describe Model)

1. Right-click anywhere on the chart background or trend line and select Describe Trend Line...
 - o This shows a quick overview of formula equations, R^2 and P-value per city.

Describe Trend Line

P-value: 0.366146
Equation: $\text{Temperature Avg (}^{\circ}\text{C)} = -0.00101407 \cdot \text{Humidity (\%)}^2 + 0.125697 \cdot \text{Humidity (\%)} + 26.1069$

Coefficients

Term	Value	StdErr	t-value	p-value
Humidity (%)^2	-0.0010141	0.0007159	-1.41646	0.157069
Humidity (%)	0.125697	0.0901763	1.3939	0.163773
intercept	26.1069	2.65435	9.83551	< 0.0001

Context Menu:

- Describe Trend Line...
- Format...
- ✓ Show Trend Lines
- Edit All Trend Lines...
- Describe Trend Model...**
- ✓ Show Recalculated Line

Copy

2. Next, right-click and select Describe Trend Model...

- o A detailed window will pop up showing the ANOVA table, Degrees of Freedom, Sum of Squared Errors (SSE), and F-statistic.

Describe Trend Model

Trend Lines Model

A polynomial trend model of degree 2 is computed for Temperature Avg ($^{\circ}\text{C}$) given Humidity (%).

Model formula: $\text{City*State}(\text{ Humidity (\%)}^2 + \text{Humidity (\%)} + \text{intercept})$

Number of modeled observations: 7310
Number of filtered observations: 0
Model degrees of freedom: 30
Residual degrees of freedom (DF): 7280
SSE (sum squared error): 259327
MSE (mean squared error): 35.6218
R-Squared: 0.0033993
Standard error: 5.9684
p-value (significance): 0.686762

Analysis of Variance:

Field	DF	SSE	MSE	F	p-value

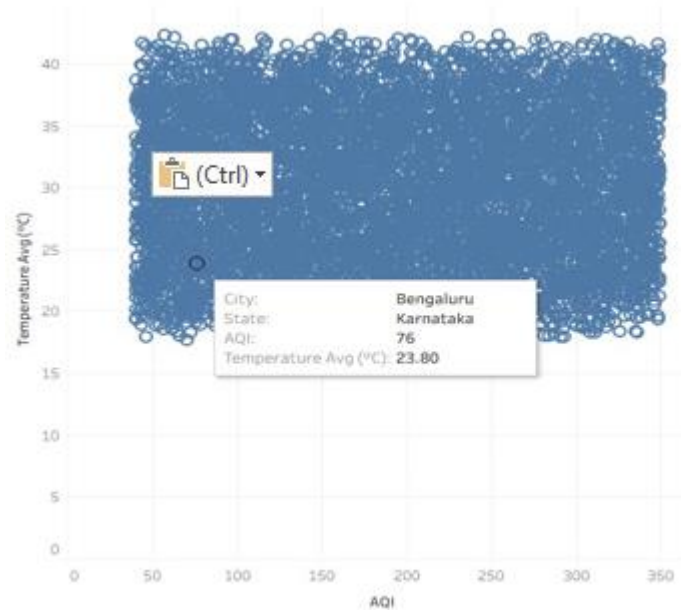
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C: Perform Clustering on Datasets :

1. Build a Multi-Metric Scatter Plot Matrix

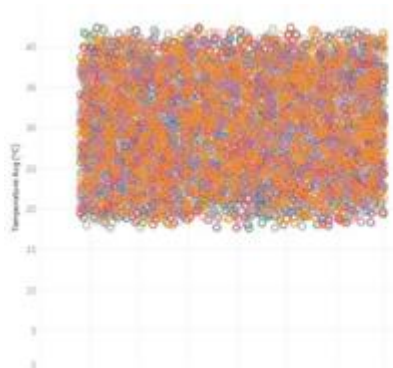
1. Open a new worksheet and name it Climate & AQI Clusters.
2. Drag AQI to Columns.
3. Drag Temperature_Avg ($^{\circ}\text{C}$) to Rows.

4. In the top menu, go to Analysis and uncheck "Aggregate Measures".
5. Drag City to Detail on the Marks card (so points represent individual city-day observations).

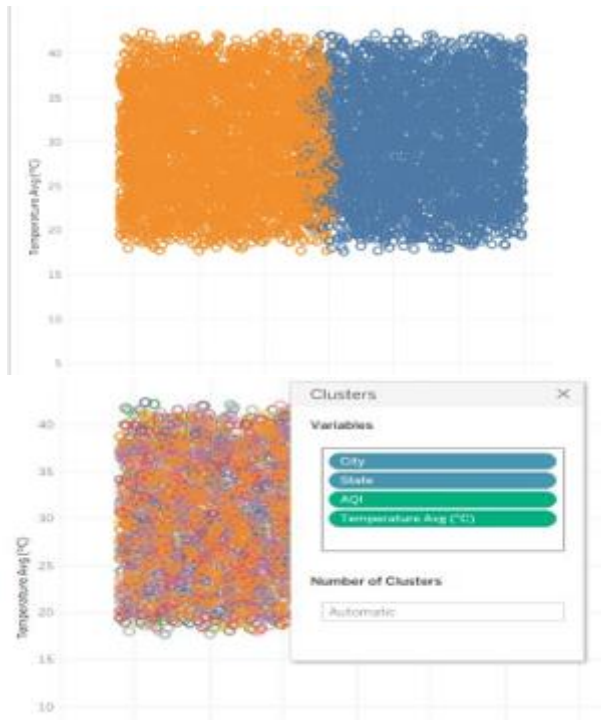


2. Apply Built-in Tableau Clustering

1. Switch to the Analytics pane on the left sidebar.
2. Under Model, drag Cluster directly onto the canvas.



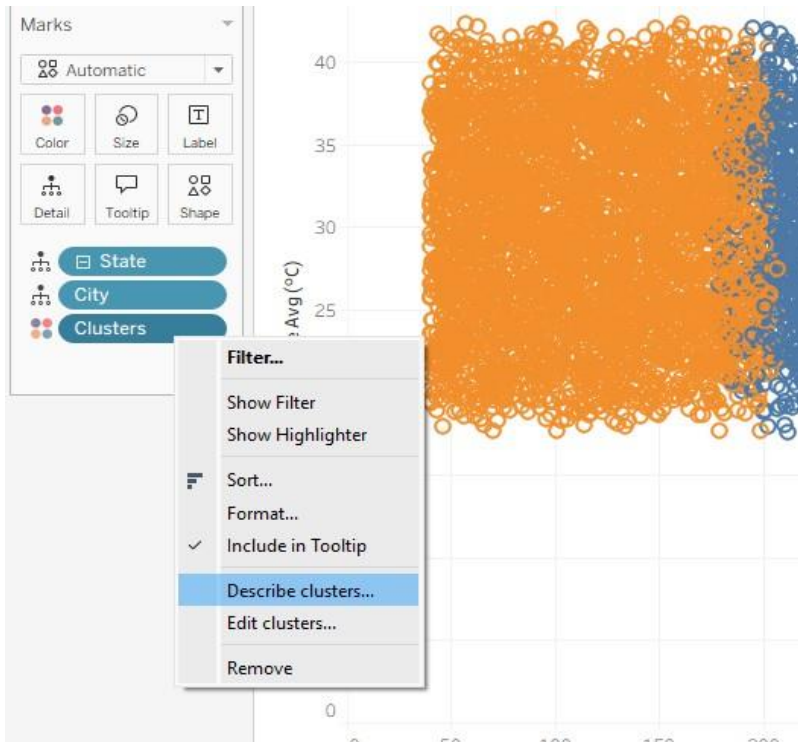
3. A Clusters dialog box will appear on the right side of the view.
4. Drag additional variables into the Variables box to create a multi-variable cluster model (If you can't find Variable box Right Click Clusters in Marks Card, click on Edit Clusters and drag and drop the variables on the box):
 - o AQI
 - o Temperature_Avg (°C)
 - o Humidity (%)
 - o Wind_Speed (km/h)



5. Under Number of Clusters, leave it as Automatic (Tableau will calculate the optimal number using K-means), or type 3 or 4 to explicitly create 3 or 4 distinct climate groups.
6. Close the cluster popup window. Tableau will color-code every data point by its assigned cluster (e.g., Cluster 1, Cluster 2, Cluster 3).

3. Analyze Cluster Profiles & Save as Dimension

1. Right-click on the Clusters field in the Marks card (or legend) and select Describe Clusters...
2. In the popup window, review two key tabs:
 - **Summary**: Shows the mean value of each metric (AQI, Temperature, Humidity, Wind Speed) for each cluster.



Describe Clusters

Summary Models

Inputs for Clustering

Variables: City, State, Sum of AQI, Sum of Humidity (%), Sum of Temperature Avg (°C), Sum of Wind Speed (km/h)

Level of Detail: City, State

Scaling: Normalized

Summary Diagnostics

Number of Clusters: 2
 Number of Points: 7310
 Between-group Sum of Squares: 459.75
 Within-group Sum of Squares: 6566.3
 Total Sum of Squares: 7026.1

Clusters	Number of Items	Sum of AQI	Sum of Humidity (%)	Centers		Sum of Wind Speed (km/h)	Most Common	
				Sum of Temperature Avg (°C)			City	State
Cluster 1	3636	271.19	61.889	30.042		13.438	Jaipur	Telangana
Cluster 2	3674	117.13	63.411	29.919		13.607	Hyderabad	West Bengal

☐ Show scaled centers

Copy to Clipboard [Learn more about the cluster summary statistics](#) Close

- o **Models:** Shows ANOVA statistics (F-statistics and p-value) indicating which variables contributed most heavily to separating the clusters.

Describe Clusters

Summary Models

Analysis of Variance:

Variable	F-statistic	p-value	Model		Error	
			Sum of Squares	DF	Sum of Squares	DF
Sum of AQI	5453.0	0.0	454.3	1	608.8	7308
Sum of Humidity (%)	12.13	0.0004987	1.002	1	603.7	7308
Sum of Wind Speed (km/h)	1.203	0.2727	0.09798	1	595.1	7308
Sum of Temperature Avg (°C)	0.7816	0.3767	0.04562	1	426.5	7308

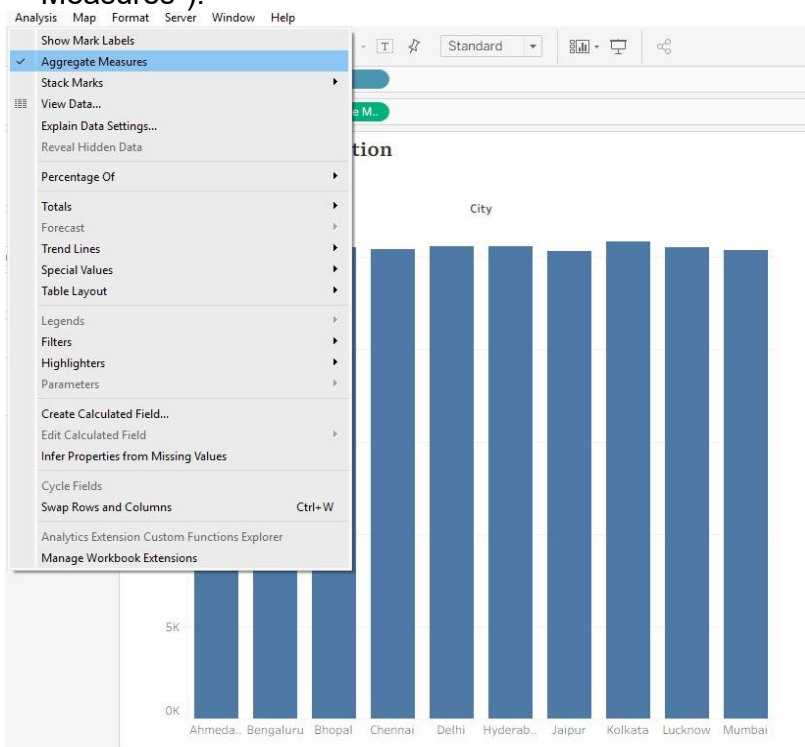
Categorical variables are not included in the Analysis of Variance table.

Copy to Clipboard [Learn more about the cluster model statistics](#) Close

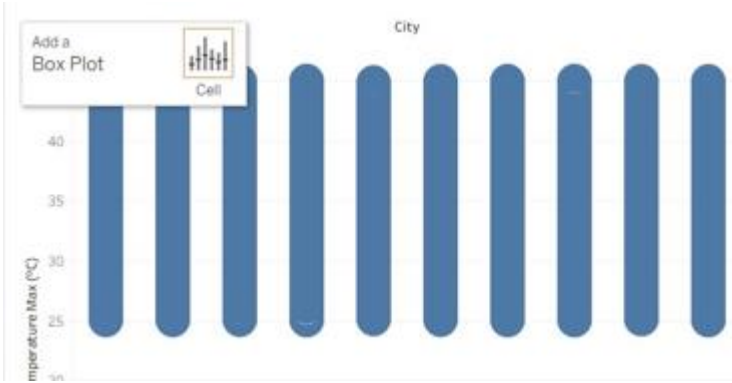
D: Analyze Distributions using Built-in Tools :

1. Create a Box Plot for Temperature Distributions by City

1. Open a new worksheet and name it Temperature Distributions.
2. Drag City to Columns.
3. Drag Temperature_Max (°C) to Rows.
4. Tableau will default to SUM(Temperature_Max). Right-click the Temperature_Max pill on Rows ☐ Measure (Sum) ☐ change it to Dimension (or go to Analysis menu and uncheck "Aggregate Measures").

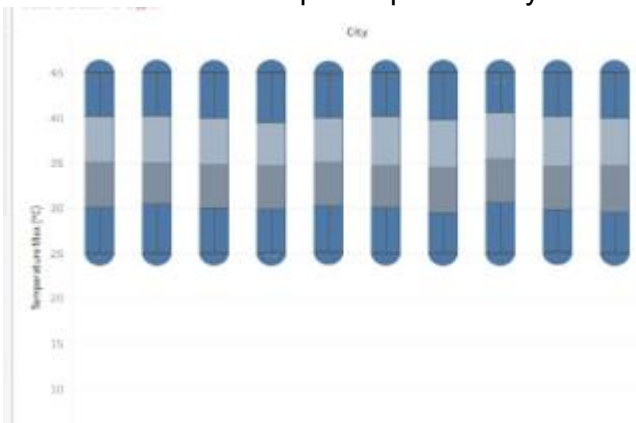


5. Switch to the Analytics pane on the left sidebar.
6. Under Custom, drag Box Plot onto the canvas and drop it on the chart.



7. Hover over any box plot to inspect key statistical metrics:

- o Upper Whisker / Maximum: Upper fence boundary.
- o Upper Hinge (Q3): 75th percentile.
- o Median (Q2): 50th percentile.
- o Lower Hinge (Q1): 25th percentile.
- o Outliers: Individual points plotted beyond the whiskers

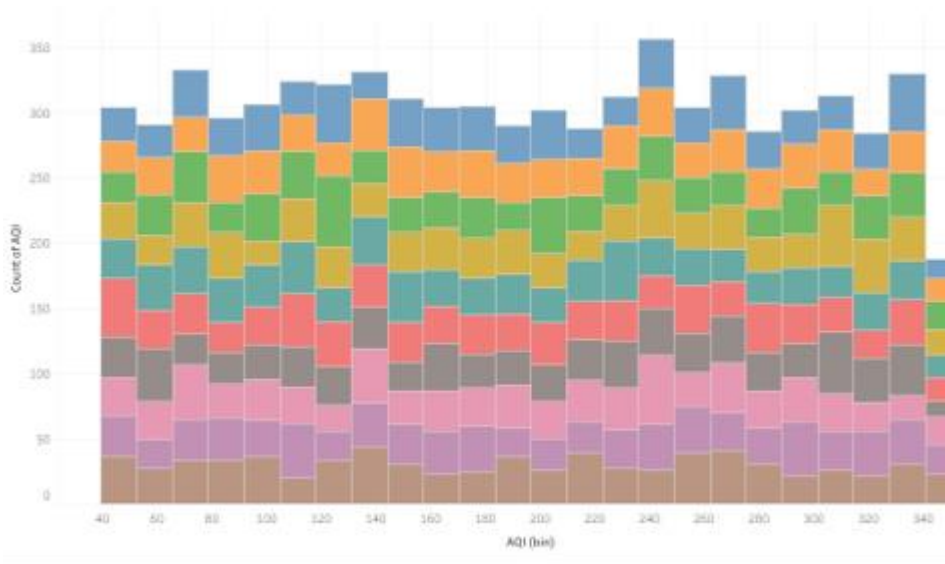


2. Build an AQI Histogram & Quantile/Percentile View

1. Open a new worksheet named AQI Histogram.
2. From the Data Pane, click on AQI.
3. Open the Show Me panel in the top-right corner of Tableau.
4. Select the Histogram icon.

(Tableau will automatically create an AQI (bin) dimension and place AQI (bin) on Columns and CNT(AQI) on Rows).

5. Drag City to Color on the Marks card to see how air quality distributions stack across cities.



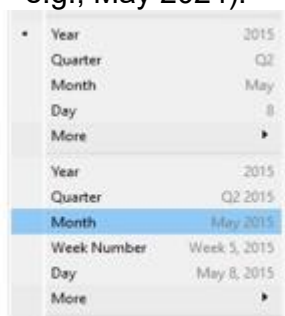
E: Apply Forecasting Techniques :

1. Set Up a Time Series Line Chart

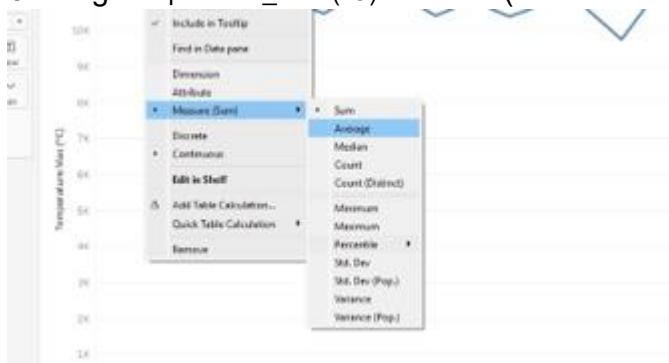
1. Open a new worksheet and name it Temperature Forecast.

2. Drag Date to Columns.

- o Note: Tableau will default to YEAR(Date). Click the dropdown arrow on the Datepill and change it to continuous Month (the second Month option in the menu, which displays as Month Year, e.g., May 2024).



3. Drag Temperature_Max (°C) to Rows (ensure it is set to AVG(Temperature_Max)).



4. Drag City to Filters and select 2 or 3 cities (e.g., Mumbai, Delhi, Bengaluru) to keep the view clean.

2. Enable Tableau Forecasting

1. Switch to the Analytics Pane on the left sidebar.
2. Under Model, drag Forecast onto the canvas and drop it.



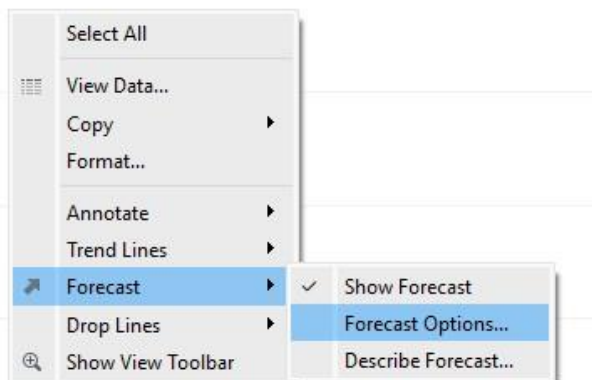
3. Tableau will instantly extend the line chart into the future with a shaded prediction interval band!

3. Customize & Analyze the Forecast

1. Right-click anywhere on the chart canvas

□ select Forecast

□Forecast Options....



2. Under Forecast Length, set it to Exactly 6 Months into the future.
3. Under Prediction Interval, verify it is set to 95%.

Forecast Options [X]

Forecast Length

☐ Automatic Next 11 months

☒ Exactly 6 Months

☐ Until 1 Years

Source Data

Aggregate by: Automatic (Months)

Ignore last: 1 Months

☐ Fill in missing values with zeroes

Forecast Model

Automatic

Automatically selects an exponential smoothing model for data that may have a trend and may have a seasonal pattern.

☒ Show prediction intervals 95%

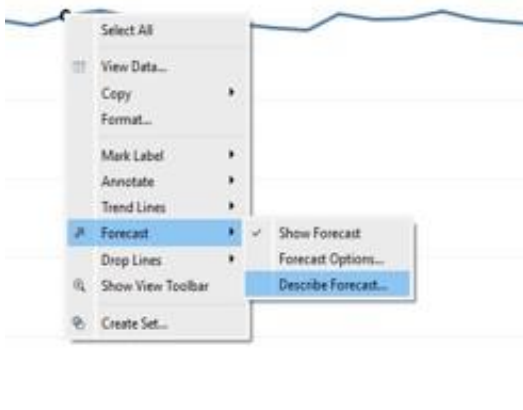
Currently using source data from January 2024 to November 2025 to create a forecast through May 2026. Looking for potential seasonal patterns every 12 Months.

[Learn more about forecast options](#)

OK

4. To evaluate the mathematical quality of the forecast:

- o Right-click the chart -> Forecast > Describe Forecast....
- o Review the Models tab to check metrics like the Root Mean Squared Error (RMSE) and whether Tableau selected an Additive or Multiplicative model for trend and seasonality.



Describe Forecast ×

Summary Models

Options Used to Create Forecasts
 Time series: Month of Date
 Measures: Avg. Temperature Max (°C)
 Forecast forward: 6 months (December 2025 – May 2026)
 Forecast based on: January 2024 – November 2025
 Ignore last: 1 month (December 2025)
 Seasonal pattern: None (Not enough data to search for a seasonal pattern recurring every 12 Months)

Avg. Temperature Max (°C)

Initial	Change From Initial	Seasonal Effect		Contribution		Quality
December 2025	December 2025 – May 2026	High	Low	Trend	Season	
35.007 ± 0.882	0.000	None		0.0%	0.0%	Ok

☐ Show values as percentages

Copy to Clipboard [Learn more about the forecast summary](#) Close

F: Compare different Statistical Models :

1. Set Up the Comparison View

1. Open a new worksheet named Model Comparison.
2. Drag Temperature_Max (°C) to Columns (set to Continuous Dimension or AVG).
3. Drag AQI to Rows (set to Continuous Dimension or AVG).
4. Drag City to Detail on the Marks card to scatter plot individual city readings.

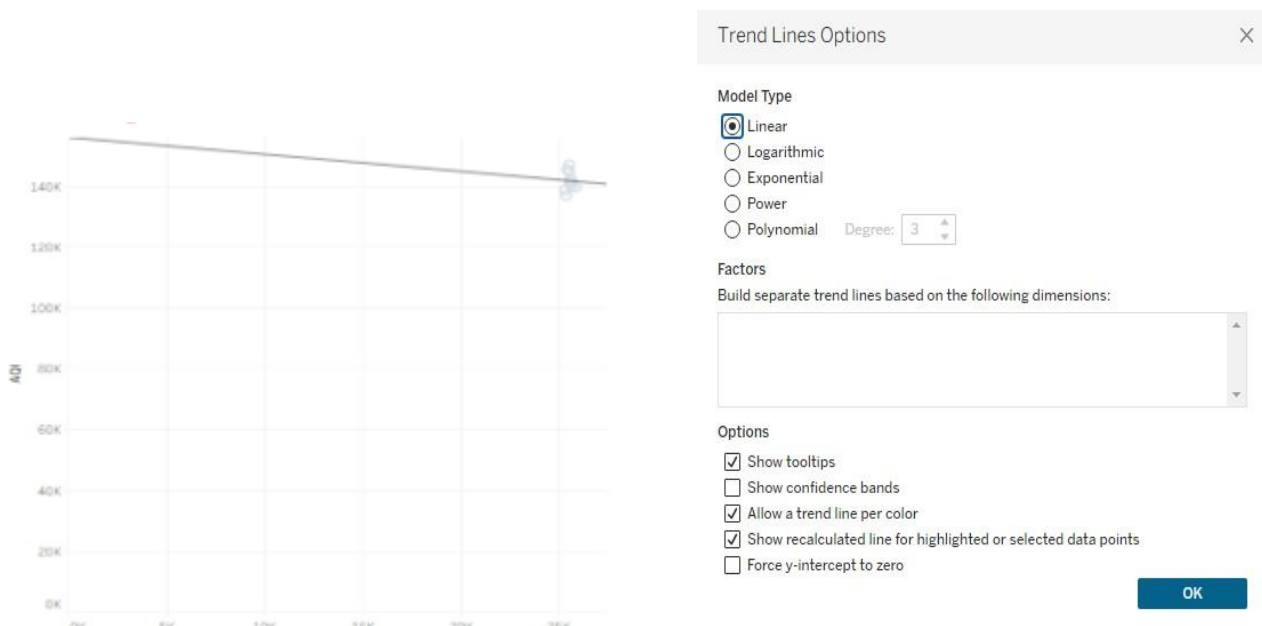
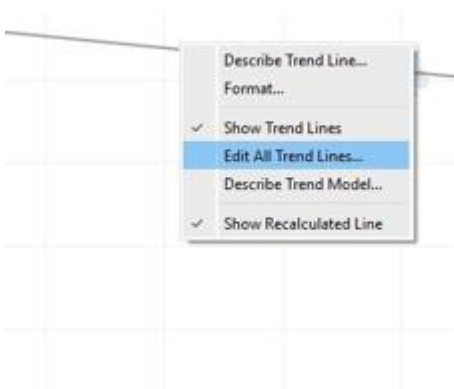


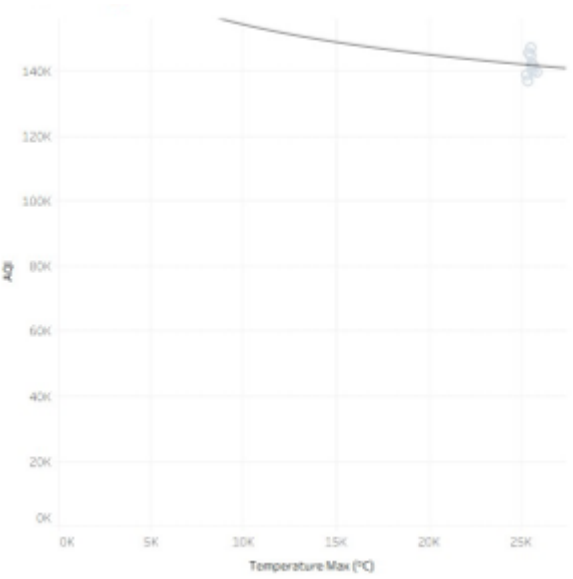
2. Add and Swap Statistical Models

1. Open the Analytics Pane on the left.
2. Drag Trend Line onto the canvas and drop it on Linear.



3. Hover over the trend line and note the R^2 and p-value.
4. Right-click the trend line □ Edit Trend Lines....
5. Switch the Model Type through the options:
 - o Linear
 - o Logarithmic
 - o Exponential
 - o Polynomial (Degree 2 or 3): Fits curved relationships.
6. Compare the R^2 value for each model type. The model with the higher R^2 and a p-value < 0.05 provides the best fit.





Trend Lines Options

Model Type

☐ Linear

☒ Logarithmic

☐ Exponential

☐ Power

☐ Polynomial

Degree: 3

Factors

Build separate trend lines based on the following dimensions:

Options

☒ Show tooltips

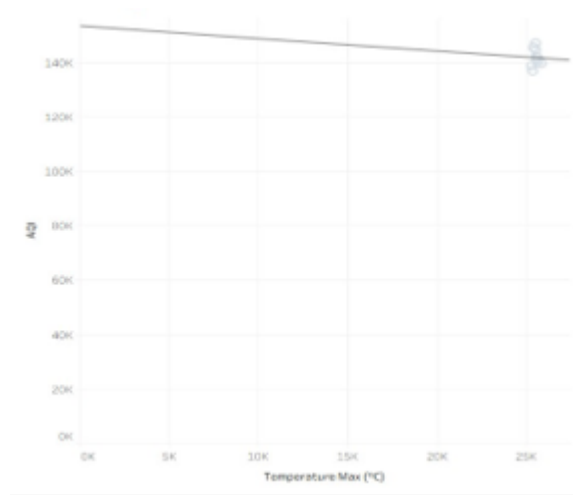
☐ Show confidence bands

☒ Allow a trend line per color

☒ Show recalculated line for highlighted or selected data points

☐ Force y-intercept to zero

OK



Trend Lines Options

Model Type

☐ Linear

☐ Logarithmic

☒ Exponential

☐ Power

☐ Polynomial

Degree: 3

Factors

Build separate trend lines based on the following dimensions:

Options

☒ Show tooltips

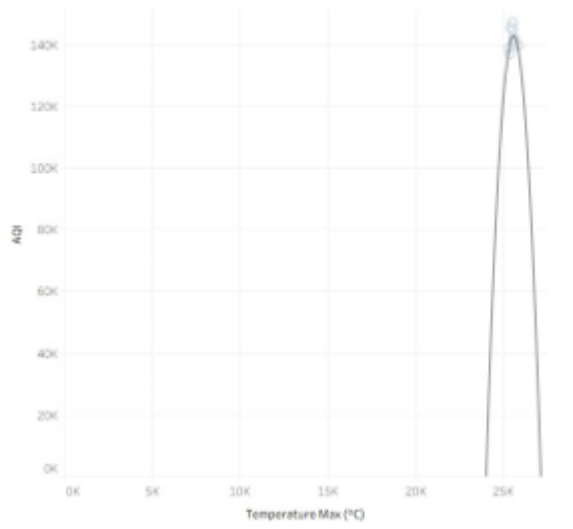
☐ Show confidence bands

☒ Allow a trend line per color

☒ Show recalculated line for highlighted or selected data points

☐ Force y-intercept to zero

OK



Trend Lines Options

Model Type

☐ Linear

☐ Logarithmic

☐ Exponential

☐ Power

☒ Polynomial

Degree: 2

Factors

Build separate trend lines based on the following dimensions:

Options

☒ Show tooltips

☐ Show confidence bands

☒ Allow a trend line per color

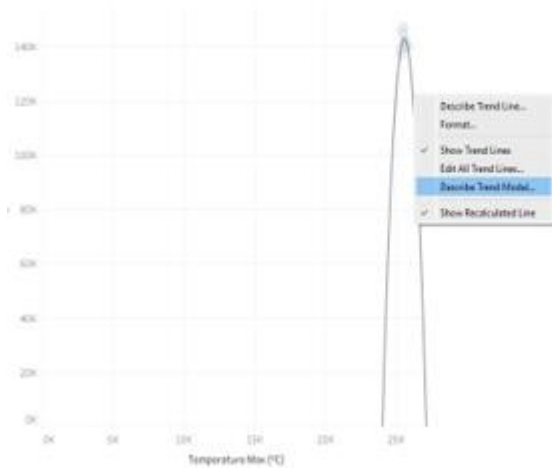
☒ Show recalculated line for highlighted or selected data points

☐ Force y-intercept to zero

OK

G: Export and Interpret Statistical Results :
1. Export the Statistical Model Description

1. On your Model Comparison worksheet (or any trend/forecast sheet), right-click the trend line or canvas.
2. Select Describe Trend Model... (or Describe Forecast... if on a forecast sheet).



Describe Trend Model

Trend Lines Model

A polynomial trend model of degree 2 is computed for sum of AQI given sum of Temperature Max (°C).

Model formula:	(Temperature Max (°C)^2 + Temperature Max (°C) + intercept)
Number of modeled observations:	10
Number of filtered observations:	0
Model degrees of freedom:	3
Residual degrees of freedom (DF):	7
SSE (sum squared error):	6.63731e+07
MSE (mean squared error):	9.48187e+06
R-Squared:	0.264967
Standard error:	3079.26
p-value (significance):	0.340466

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